



CortexOS

Capabilities Manual

Jobs, skills, triggers, workflows, human choices, and evidence

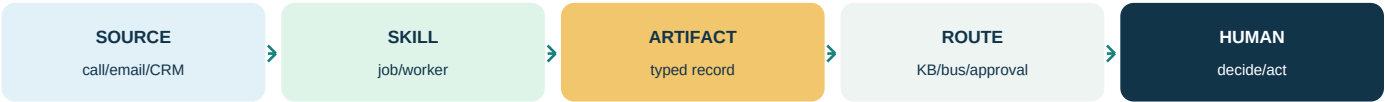
This manual answers:

What happens automatically? What do I ask for? Where does the result land? How do I know it really happened?

Built from the current repository, knowledge-sync system guide, meeting-chain audit, and independent verifier reports. Altari source: [knowledge-sync/raw/areas/clearworks/altari-skilltree](#) (PHASE-1-WIRING-PLAN, job playbooks, DESIGN-A-G). Code exists is never presented as live output.

1. The operating model

One source event enters a specialist job, creates a typed artifact, and routes to knowledge, the bus, Multica, or an approval surface.



Mode	Meaning	Human expectation
AUTOMATIC	Cron, event, daemon hook, or bus mutation invokes it.	You receive a structured artifact or exception without remembering to ask.
MANUAL	You explicitly invoke or approve it.	Use for judgment, sensitive data, one-offs, and deliberate re-runs.
HYBRID	Automation prepares; a human approves.	The system does repetition; you decide consequence.
UNVERIFIED	Code/config exists but runtime/output is not proven.	Available-but-not-promised until the live probe succeeds.

2. P1-P6 foundation and operations

The foundation makes the fleet searchable, observable, safe, and reviewable.

Phase / job	What you get	Manual choice	Automatic trigger
P1 knowledge foundation	Router, org brain, portal/memory/session fold-ins; searchable source-backed context.	Ask for kb-query, an import, or a deliberate backfill.	Router called by output jobs; reconcile nightly; source sync jobs (some need live probe).
P2 contract rollout	Skills share an I/O/provenance contract; lint catches drift.	Run contract-lint before accepting a skill.	Build/CI gate; declared scope complete.
P3 Multica rail	Agent work becomes trackable issues; status can return to bus.	Run bounded dry-run, then approve a real push.	Bus create/update/claim/complete mirrors outbound; inbound and fresh runtime need proof.
P4 signals + approvals	Blocked/due/quiet exceptions and human approval surface.	Run a dry signal sweep or review approvals.	Signal-sweep cron and job-created approvals.
P5 cron + event hygiene	Dead/duplicate cron drift is visible; events can replace polling after proof.	Request fleet audit; choose event cutover.	Cron registry/migration and event receivers; never kill a poll before equivalent output.
P6 weekly review	A repeatable findings, fixes, and decisions packet.	Say: run weekly review.	Weekly cadence; four consecutive fires earn the cadence.

3. Loops, waves, daemon, reliability

Continuity machinery protects user-facing jobs from context loss, duplicate execution, and silent cron drift.

Loop / wave	Purpose	Human action	Automatic behavior
LOOP1 / Wave 1	Real-window context handoff and continuity.	Approve live-promote/restart after evidence.	Daemon handoff/restart machinery.
LOOP2	Resolve retrieval-enforcer fork delta.	Choose delete vs keep-with-bypass.	No auto-completion; file still exists.
LOOP3-5 / Wave 2	Keep upstream continuity/PTY/runtime primitives; converge wounds.	No routine action.	Daemon/PTY/runtime behavior.
LOOP6-7	Cron-active marker and inbound persistence.	Probe after restart.	AgentManager and conversation-buffer writes.
Wave A/B	Completion only after true verification; re-earn receipts.	Review evidence receipts.	Pipeline/ledger gates.

Daemon jobs include cron scheduling/migration, context-handoff leases, crash/stall detection, single-flight restart protection, PTY lifecycle, inbound persistence, heartbeats, and runtime drift checks. You choose a restart/live-promote; the daemon performs routine scheduling and recovery.

4. Altari capabilities

Human-shaped jobs: executive assistance, CRM hygiene, meeting intelligence, KB maintenance, scoping, delivery, reliability, and the Mission Control surface.

Capability	Human outcome	Manual choice	Automatic trigger
A executive assistant / inbox / booking	Triage, briefs, booking and Zoom coordination; drafts for review.	Ask for triage, a booking search, or a draft refresh.	PA comms-check, morning brief, evening wrap; Gmail/Zcal deployment varies.
B enrichment + records admin	Sourced firmographics and evidence-backed CRM change list.	Enrich one account; review/approve a reconciliation batch.	CRM event lane plus weekly backstop intended; never auto-apply live records.
C meeting intelligence + followup	A call becomes outcomes, actions, decisions, deal-state, files, and recap.	Process pasted notes or deliberately re-run one meeting.	Fireflies login/registration is done and a real HMAC event reached PA; downstream PA writers and single-owner runtime proof remain.
D KB maintenance	Dead links, duplicates, stale/orphan content and retrieval findings.	Run focused hygiene before a migration.	Nightly deterministic + weekly verify + event stale-marking.
E integration/scoping gate	Complexity/price cannot finalize without API reality and exemplar grounding.	Ask auditmaster to scope an integration.	Mandatory pre-scope pipeline gate.
F delivery status	Evidence-backed per-client status draft and decisions needed.	Ask for an on-demand client packet.	Weekly draft to approval queue; never auto-send.
G reliability / production stack	Daily stall inventory catches silent client-facing failures.	Ask for a production-stack audit after deploy.	Daily stall-detection cron.
Mission Control v2	KPIs, client phases, signals, approvals, activity and fleet ops in one room.	Open it to decide what deserves attention next.	Dynamic route and API snapshot; runtime/listener freshness needs proof.

The scope below is read from the Altari folder's PHASE-1-WIRING-PLAN and DESIGN-A through DESIGN-G. It names the jobs that were actually incorporated or deliberately evaluated; it does not treat the full 137-job catalog as wired.

5. Altari scope: job → skill → operating lane

These are the incorporated Altari jobs and the associated skills we actually read. The skill's original Altari job playbook supplies the human outcome; the CortextOS I/O contract supplies the safe destination and approval rule.

Altari workstream	Incorporated job(s)	Skill(s) actually used	How it is used in CortextOS
A · proactive EA	Calendar Management; Inbox Triage; Meeting Booking	executive-assistant + inbox-manager + meeting-booking-coordinator (absorbed into PA comms-check)	Event classifier routes booking/inbox signals; human invokes an on-demand booking or triage run. Gmail drafts and calendar holds stay approval-gated.
B · CRM	Account Enrichment; Contact Enrichment; Email Verification; CRM Sync; Entity Compliance; Document Filing & Retrieval	data-enrichment-specialist + records-administrator	CRM receives evidence-backed enrichment and a diff. Records admin prepares sync/compliance/filing work; merges, deletes, filings, and writes remain human-approved.
C · meeting suite	Call Capture; Transcript Processing; Post-Call Debrief; Meeting Follow-Ups; Follow-Up Drafting	meeting-intelligence-engineer + deal-debrief-analyst + followup-coordinator	Fireflies/notes become a client-scoped meeting record, decisions, open items, CRM delta, task/approval, and recap draft. PA is the canonical owner.
D · knowledge	Context Maintenance; Company Knowledge Base	knowledge-base	Per-client context and unified corpus are reconciled, deduped, and made searchable. Scheduled maintenance is useful only after freshness/output probes pass.
E · scoping + delivery	Integration Builds; Data Migration; Demo Prototyping; Deal Room Assembly; Status Updates	integration-engineer + solutions-engineer + deal-room-producer + delivery-status-reporter	Auditmaster uses integration and exemplar gates before scope/price; deal room and prototype are on-demand; delivery reports draft good news and route bad news to a human.
G · reliability decision	QA & Verification; Monitoring/Alerting; Incident Response	native verify/true-verify/red-team/sage/sre/codeburn cover most of this; no separate Altari worker	Keep the native gates. Add an incident-response runbook to SRE only if the live operating review shows a gap.

The Altari ladder is preserved: human-led → human-assisted → autonomous. CortextOS currently promises the assisted rung unless a live event, durable artifact, and independent re-run prove the autonomous rung.

6. Altari job cards: how each one is used

Use this as the practical index. Every row answers what starts the job, what it writes, where it goes, and what still belongs to you.

Job → associated skill	Use it when / input	Output and destination	Auto trigger / human gate
Call Capture → meeting-intelligence-engineer	Every recorded call, Fireflies event, or pasted notes; client + meeting ID.	Transcript/meeting note with provenance; canonical meetings + client file.	Fireflies webhook or PA poll. Human reviews only sensitive/do-not-record cases.
Transcript Processing → meeting-intelligence-engineer	After capture; raw transcript or notes.	Structured facts, decisions, actions, open questions; searchable transcript archive.	Same meeting chain. No orphan transcript; do-not-record stores summary only.
Post-Call Debrief → deal-debrief-analyst	A sales/discovery call where outcomes or deal state changed.	Approval-ready CRM delta and concise debrief.	Event/poll candidate; human approves consequential CRM changes.
Meeting Follow-Ups → followup-coordinator	An explicit client or internal action was agreed.	Owner/date/action rows, task/Multica or approval route, recap input.	Meeting extraction. Do not create a task from casual self-talk.
Follow-Up Drafting → deal-debrief-analyst	A real follow-up is owed after a call.	Same-hour Gmail draft: decisions, next steps, owners/dates, one forward-looking line.	PA recap worker after event/poll; human edits and sends.
Pre-Call Briefing → call-prep-researcher	Upcoming external meeting; calendar attendee + CRM/history/email context.	One-page cited brief: who, history, open threads, objectives.	Calendar cron/weekday page. Human uses it before the call; attendee parsing must be verified.
Inbox Triage → inbox-manager / inbound-response-manager	Unread/recent threads or pasted batch.	NEEDS YOU / QUICK REPLY / WAITING / FYI / REVIEW digest; drafts only.	PA comms-check. Human approves every send; never delete/archive/commit.
Calendar + Meeting Booking → executive-assistant / meeting-booking-coordinator	Interested reply plus duration, timezone, and constraints.	2–3 concrete slots in both timezones; tracker state; reminder/recovery drafts.	Comms event or on-demand PA entry. Human confirms calendar invite; max two no-show touches.
Enrichment + Verification → data-enrichment-specialist	Scored account/contact that needs context; domain or person identifiers.	Provenance-tagged firmographics, contacts, email status, confidence/cost notes.	CRM lane/backstop intended. Nothing UNVERIFIED becomes sendable; optional Apollo/Exa/verifier keys.
CRM Sync + Filing/Compliance → records-administrator	CRM/activity diff, documents, or entity obligation register.	MATCH/STALE/MISSING/DUPLICATE/ORPHAN diff; filed/indexed docs; 30/14/3-day alerts.	Records sweep intended weekly/config-only today. Human approves merges/deletes and signs filings.
Context Maintenance → knowledge-base	After meeting, email, deliverable, or scheduled corpus check.	Updated client STATE/context, dedupe/reconcile findings, searchable index receipt.	Nightly/weekly KB jobs. Human resolves ambiguous/stale facts.
Integration + Migration → integration-engineer	Approved flow with real source/destination and field map.	Idempotent implementation, mapping, retry/dead-letter log, row-count reconciliation.	Mandatory scoping gate; human signs field map and tests real data before handoff.
Prototype + Deal Room → solutions-engineer / deal-room-producer	Discovery transcript or approved proposal package.	Same-day React/Mermaid/Notion prototype + send draft; deal room with proof/pricing/one next step.	On-demand after discovery. Human chooses what to polish, share, and send.
Status Updates → delivery-status-reporter	Active client cadence or shipped milestone.	Good/neutral client draft; bad/mixed private brief with impact/options/recovery plan.	Weekly cadence/delivery event. Good news can automate after approval; bad news always human-first.

7. Meeting filters: commitment, not chatter

The filter is intentionally conservative. A future-tense first-person sentence is not automatically a task. It must describe a concrete work item or decision, have a resolvable owner, and have a usable due signal; otherwise it remains context or is dropped with an auditable reason.

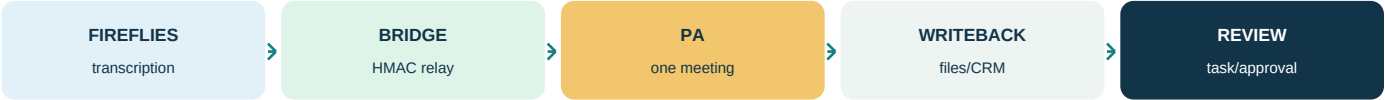


Transcript text	Classification	Expected behavior
"I'll send the routing file by Friday."	ACTIONABLE	Create an open item only if speaker/owner is known; due date Friday; preserve quote.
"We should revisit pricing next Tuesday."	ACTIONABLE / VERIFY	Create a proposed follow-up with date; do not claim a firm commitment if wording is tentative.
"I'll be learning more about coding in the next few weeks."	CONTEXT / CASUAL	No task, no owner, no CRM change, no Multica item. Record only as context if relevant.
"I'm going to think about it."	CONTEXT / NON-COMMITMENT	No task; ask for an explicit next step only if the meeting requires one.
"Sarah will send the contract tomorrow."	ACTIONABLE	Owner=Sarah, due=tomorrow (resolved to date), source quote retained.
"Please don't record this part."	DO-NOT-RECORD	Store only an allowed summary; never persist verbatim transcript or extract tasks from the flagged segment.

Acceptance set: test named owners, “I / I’ll / I’m going to / we” variants, vague timeframes (“a day or so”), multiple meetings with attached child rows, duplicate CRM contacts, casual no-yield speech, and do-not-record segments. A filter that turns “I’ll be learning more about coding in the next few weeks” into a task is failing the product requirement.

8. The complete meeting workflow

The webhook is real. The present gap is downstream persistence: PA must fetch the single meeting by ID and invoke the durable writers before polling is retired.



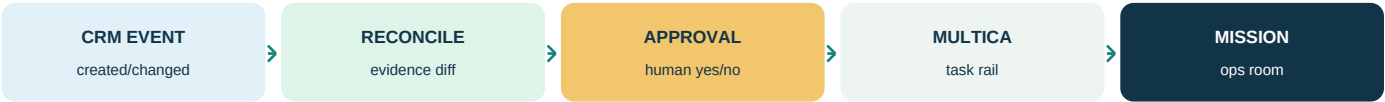
Stage	What should happen	Current evidence / acceptance
1 Capture	Fireflies sends transcription-complete; bridge verifies and relays.	PROVEN: real 01KZ71M... event arrived 2026-08-04 23:17Z.
2 Extract	PA honors FF_MEETING_ID; actions, decisions, deal-state, zero-yield reason.	Two decisions extracted; current PA worker still needs meeting-ID proof.
3 File	Transcript and note land in canonical paths with provenance.	Writeback invoker absent; not proven.
4 Update	Client History/Open Items adds dated rows; unknowns stay NEEDS-OWNER.	Merge logic needs source-aware tests.
5 CRM/Rail	One deterministic CRM event; agent work Multica; human work approval.	Producer/invokers incomplete; keep polling.
6 Recap	Draft includes tracked items and one next step.	Scheduler fired historically; current PA draft output needs proof.

Meeting output sample

OUTCOMES: ops-line funded; rollout scoped
DECISIONS: no procurement review; three-region target
OPEN: routing volumes | Sarah Chen | 2026-08-06
OPEN: SLA language | NEEDS-OWNER | NEEDS-DEADLINE
NEXT: proposal review | 2026-08-18

9. CRM, knowledge, and Mission Control

This is how the broader system feels to the human, not just how a script is invoked.



Surface	Example output	Interpretation
Meeting file	OUTCOMES: 3 bullets; DECISIONS: 2 facts; DEAL-STATE: pilot -> rollout	The call is durable/searchable.
Client Open Items	Send routing volumes Sarah Chen 2026-08-06 OPEN Revisit SLA language NEEDS-OWNER NEEDS-DEADLINE OPEN	Unknowns remain visible; no invented owner.
Approval queue	CRM reconciliation: 2 STALE, 1 MISSING, 0 auto-applied	You choose whether records change.
Mission Control	5 KPIs; 2 at-risk signals; 1 approval; 4 open items	Attention surface, not decorative status.
Weekly review	FINDING: stale sync; FIX TASK: bounded probe; DECISION: approve org	Every finding becomes work or a decision.
Zero-yield ledger	meeting_id=... drop_reason=casual	No-output is auditable, not silently burned.

Knowledge hygiene criteria

- canonical content-type home
- source/transcript/job provenance
- NEEDS-OWNER and NEEDS-DEADLINE instead of guesses
- rebuilds preserve meeting-derived history but do not immortalize stale CRM rows
- live changes produce an approval list
- zero-output reasons are auditable.

10. When you invoke vs when the system fires

Use these prompts as the memory aid. Automatic jobs should leave evidence; manual actions supply judgment, scope, or permission.

Human says / event	What runs	Expected result
Process my latest call	PA meeting intelligence + writeback + recap chain.	Meeting/client records, owner/deadline table, recap draft, routed task/approval.
What is open with [client]?	Followup coordinator reads client Open Items and deadlines.	Open rows, overdue chase, source references.
Run CRM reconciliation	Records administrator builds A/B evidence diff.	MATCH/STALE/MISSING/DUPLICATE/ORPHAN list; no silent writes.
Show delivery status	Delivery reporter composes draft.	Client-by-client progress, risks, shipped work, decisions needed.
Run weekly review	P6 deterministic review aggregates findings.	Findings, fix tasks, cadence receipt, human decisions.
Fireflies transcription-complete	Bridge verifies HMAC and relays to PA.	Single-meeting extraction; durable writes only once the invokers are proven.
Bus task mutation	Multica mapping/push + sync receipt.	Issue and bus state align; fresh external proof required.
Nightly/weekly cron	KB, reliability, CRM, or review job.	Structured finding/digest; silent OK if no action.

Typical day: meetings and inboxes arrive automatically; KB/reliability jobs keep the index healthy; Mission Control collects exceptions; you review approvals and decisions; manual invocations handle one-off research, sensitive records, backfills, and deliberate re-runs.

11. What remains

The verifier loop found bounded work packages; these are the actual done conditions.

Order	Work package	Proof of done
1	Cron migration fail-closed preservation	Normal merge + corrupt-registry tests pass; runtime crons survive restart.
2	PA canonical meeting lane	PA honors meeting ID; writeback/recap reproducible; real event creates durable files.
3	FR-001 through FR-005	Binding cases, traveling tests, runtime-policy evidence, staged files pass independent review.
4	Deterministic CRM persistence	One real meeting ID yields exactly one interaction/followup/meeting artifact.
5	Multica + Mission Control	Fresh sync/issue proof and dashboard snapshot after runtime restart.
6	LOOP2 + runtime promotion	Delete-vs-bypass decision recorded; daemon directly probed; duplicates retired only then.

Evidence rule: each lane gets an isolated worktree/branch, independent maximum-reasoning review, focused tests, and a safe real-life probe wherever possible. A green build alone never proves a runtime claim.